

Ex: 2b**VM-VM ping, VM-Native Ping****Date: 7/9/2026**

Aim:

To check the target is alive between vm-vm and vm-native using the ping command .

Steps: Before ping commands

- Both VMs are Powered ON
- Both are connected to the same network (e.g., Host-Only or Bridged)
- Each VM has a valid IP address (like 192.168.56.x)
- Firewall is disabled or ICMP is allowed

Find IP Addresses

- 1) Using ip a or ifconfig in VM
- 2) Using ipconfig in windows

To enable VM-to-VM and VM-to-host ping communication in VirtualBox, first set both VMs' network adapters to "Host-Only Adapter" (e.g., vboxnet0) via VirtualBox Settings Network. Then, start both VMs and run ip inside each to find their IP addresses (usually 192.168.56.x). On each VM, disable the firewall using `sudo ufw disable` and optionally flush iptables with `sudo iptables -F`. On the Windows host, enable ICMP ping by opening "Windows Defender Firewall with Advanced Security", going to Inbound Rules, and enabling "File and Printer Sharing (EchoRequest-ICMPv4-In)". Now, use `ping <other_VM_IP>` from one VM to the other, and from the host, use Command Prompt to ping each VM's IP (e.g., `ping 192.168.56.101`)

1)VM – NATIVE:

To check the target is alive between the host and vm using ping commands

IP address of windows: 10.10.32.103

IP address of ubuntu: 172.17.0.1

IP address of kali: 10.10.2.15

```

C:\ Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::111c:7d06:f293:f647%15
    IPv4 Address. . . . . : 10.10.32.103
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.10.32.1

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::1dbc:8b5c:6841:65de%26
    IPv4 Address. . . . . : 192.168.142.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

```

Ping in windows

```

C:\ Command Prompt

C:\Users\Admin>ping 172.17.0.1

Pinging 172.17.0.1 with 32 bytes of data:
Reply from 172.17.0.1: bytes=32 time<1ms TTL=64
Reply from 172.17.0.1: bytes=32 time<1ms TTL=64
Reply from 172.17.0.1: bytes=32 time<1ms TTL=64
Reply from 172.17.0.1: bytes=32 time<1ms TTL=64

Ping statistics for 172.17.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Admin>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

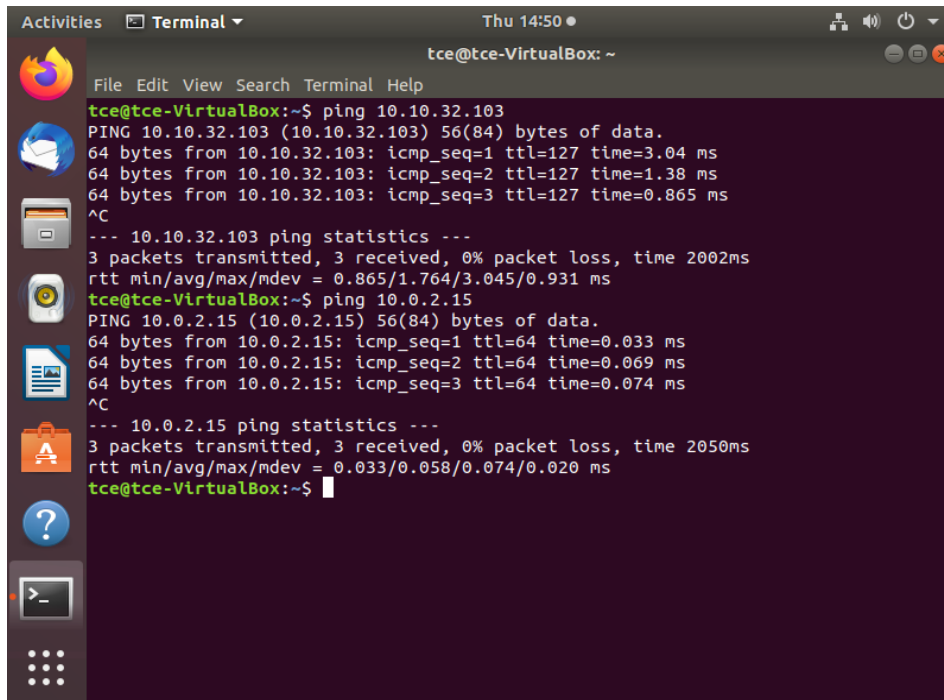
Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

VM- VM

I am using linux (one was ubuntu) and another was (Kali linux)

Ping in Linux ubuntu:

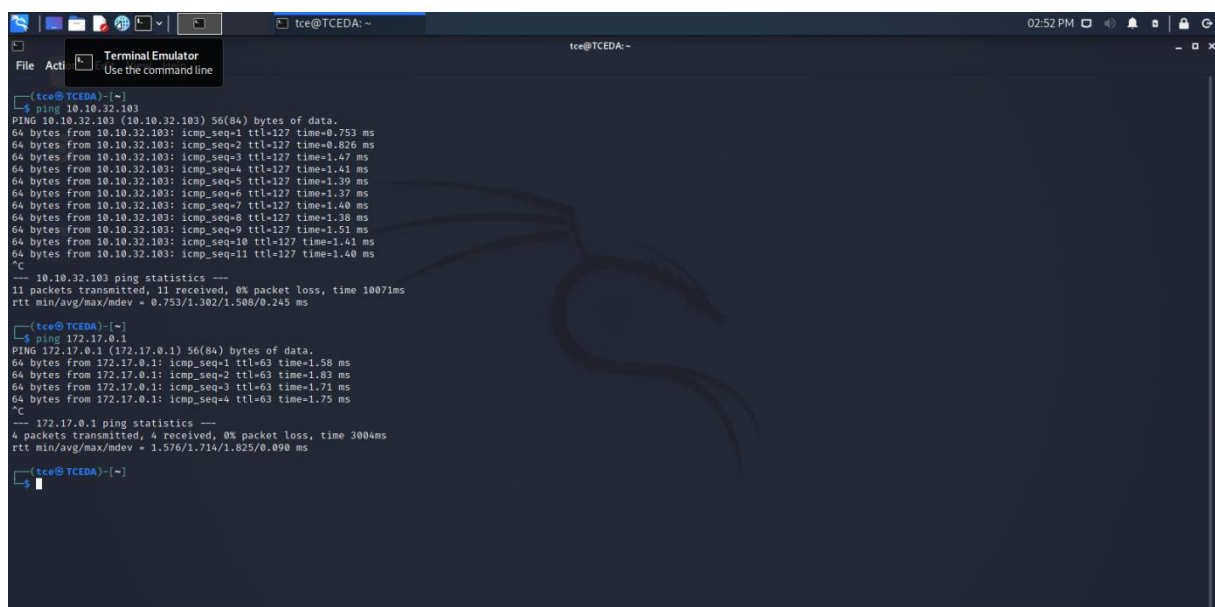


```

Activities  Terminal  Thu 14:50
tce@tce-VirtualBox: ~
File Edit View Search Terminal Help
tce@tce-VirtualBox:~$ ping 10.10.32.103
PING 10.10.32.103 (10.10.32.103) 56(84) bytes of data.
64 bytes from 10.10.32.103: icmp_seq=1 ttl=127 time=3.04 ms
64 bytes from 10.10.32.103: icmp_seq=2 ttl=127 time=1.38 ms
64 bytes from 10.10.32.103: icmp_seq=3 ttl=127 time=0.865 ms
^C
--- 10.10.32.103 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 0.865/1.764/3.045/0.931 ms
tce@tce-VirtualBox:~$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.033 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.069 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.074 ms
^C
--- 10.0.2.15 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2050ms
rtt min/avg/max/mdev = 0.033/0.058/0.074/0.020 ms
tce@tce-VirtualBox:~$

```

Ping in kali Linux:



```

tce@TCEDA: ~
File Acti Terminal Emulator
Use the command line
tce@TCEDA: ~
tce@TCEDA:~$ ping 10.10.32.103
PING 10.10.32.103 (10.10.32.103) 56(84) bytes of data.
64 bytes from 10.10.32.103: icmp_seq=1 ttl=127 time=0.753 ms
64 bytes from 10.10.32.103: icmp_seq=2 ttl=127 time=0.826 ms
64 bytes from 10.10.32.103: icmp_seq=3 ttl=127 time=1.47 ms
64 bytes from 10.10.32.103: icmp_seq=4 ttl=127 time=1.41 ms
64 bytes from 10.10.32.103: icmp_seq=5 ttl=127 time=1.39 ms
64 bytes from 10.10.32.103: icmp_seq=6 ttl=127 time=1.37 ms
64 bytes from 10.10.32.103: icmp_seq=7 ttl=127 time=1.40 ms
64 bytes from 10.10.32.103: icmp_seq=8 ttl=127 time=1.38 ms
64 bytes from 10.10.32.103: icmp_seq=9 ttl=127 time=1.51 ms
64 bytes from 10.10.32.103: icmp_seq=10 ttl=127 time=1.41 ms
64 bytes from 10.10.32.103: icmp_seq=11 ttl=127 time=1.40 ms
^C
--- 10.10.32.103 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 1007ms
rtt min/avg/max/mdev = 0.753/1.302/1.506/0.245 ms
tce@TCEDA:~$ ping 172.17.0.1
PING 172.17.0.1 (172.17.0.1) 56(84) bytes of data.
64 bytes from 172.17.0.1: icmp_seq=1 ttl=63 time=1.58 ms
64 bytes from 172.17.0.1: icmp_seq=2 ttl=63 time=1.83 ms
64 bytes from 172.17.0.1: icmp_seq=3 ttl=63 time=1.71 ms
64 bytes from 172.17.0.1: icmp_seq=4 ttl=63 time=1.75 ms
^C
--- 172.17.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.576/1.714/1.825/0.090 ms
tce@TCEDA:~$

```

RESULT

Thus checking the victim or target is alive between vm-vm and vm-native using the ping command has been executed successfully and output has been verified.